

Assignment 4 — Poori CRUD API Banao

Concept: REST methods, status codes, PATCH vs PUT, findById

Level	Medium
Time	1.5 hours
Zaroori	Assignment 1, 2, 3

Kya seekhoge

- Kaunse operation pe kaunsa HTTP method
- 200 vs 201 vs 404 kab
- PATCH aur PUT me asli fark
- .save() aur findByIdAndUpdate() me kya farak padta hai

Problem

Review ke liye 5 APIs banao — poore layer structure ke saath (route → validation → controller → service).

#	Method	Route	Kaam	Success code
1	POST	/reviews/createReview	Naya review	201
2	GET	/reviews/getReviews	List + filter + pagination	200
3	GET	/reviews/getSingleReview/:id	Ek review	200
4	PATCH	/reviews/updateReview/:id	Update	200
5	DELETE	/reviews/deleteReview/:id	Delete	200

Requirements

- :id wale teeno routes me reviewIdSchema se params validate karo. Galat ObjectId pe DB call jaani hi nahi chahiye.
- Review nahi mila to 404 — aur ye decision service me hona chahiye, controller me nahi:

```
js if (!review) throw ...  
// 404
```
- Update me .save() use karo, findByIdAndUpdate nahi. Assignment ke saath likh ke bhejo ki kyun.
- Delete me findByIdAndDelete use karo. Wo null deta hai agar item nahi mila — isse extra findById bachegi.
- getReviews me ye 3 cheezein chalein: - ?status=approved - ?minRating=4 - ?page=2&limit=5

Test karne ki list

Request	Expected
POST sahi data	201
POST rating: 10	400
GET /getSingleReview/123	400 (galat ObjectId)
GET /getSingleReview/<sahi format, exist nahi>	404
PATCH khaali body {}	400
PATCH { "rating": 4 }	200, baaki fields waise ki waise rahein
DELETE ek baar	200
DELETE wahi dobara	404

Aakhri wala test bahut important hai. Delete karke dobara delete karo — agar 200 aa raha hai to tumhara code bug wala hai.

Soch ke batao

1. `PUT` use karke `{ "rating": 4 }` bhejo — `title` aur `comment` ka kya hoga? `PATCH` me kya hoga? Ek line me fark likho.
 2. Create pe `201` kyun, `200` kyun nahi? Frontend ko isse kya fayda?
 3. Delete safal hone pe deleted review wapas bhejna chahiye ya nahi? Apni raay do.
 4. `findByIdAndUpdate` me validators by default chalte hain ya nahi? Nahi chalte to chalane ke liye kya likhna padta hai?
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Bonus

Ek **6th API** banao: `PATCH /reviews/:id/approve`

- Sirf `status` ko `"approved"` kare
- Agar review pehle se approved hai to `400` de ke bata de ki *"Review is already approved"*

Socho — ye alag API kyun banayi, `updateReview` se hi kyun nahi kar diya?